Owen Shi

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EDUCATION

Harvard College **Cambridge, MA**
B.S in Electrical Engineering, Minor in Computer Science | Expected May 2030 GPA: 3.8 (Predicted)

- Harvard Undergraduate Aerospace Collective (Rocketry), Harvard Undergraduate Robotics Club

Stuyvesant High School **Manhattan, NY**
Stuyvesant Advanced Regents Diploma SAT: 1570 (800 Math) | GPA: 97.45

WORK & LEADERSHIP EXPERIENCE

Stuy Fusion, FTC 479 **Manhattan, NY**
Vice President *Sept. 2023 - July 2026*

- Engineered an **AprilTag** computer vision pipeline for field localization, odometry, and autonomous alignment.
- Implemented multithreaded robot controls, boosting loop frequency to **10 kHz** for low-latency response times.
- Migrated codebases to a command-based architecture, improving modularity, testing, and subsystem reuse.
- Developed and deployed the team's website to strengthen digital outreach, recruiting, and sponsor engagement.
- Designed beginner education workshops and delegated technical projects across ~**32 members**.

BCTCio & Future Sphere Education **Remote**
Intern Full-Stack Web Developer *Feb. 2022 - Sept. 2025*

- Delivered full-stack features for freelanced web apps using modern frameworks, databases, APIs, and cloud tools.
- Maintained a reusable UI npm package with **30+ production-ready components** for consistent client development.
- Created and maintained Future Sphere Education's student portal**,** serving **4,000+ active student users**, **300+ classes**, and **12+ teachers**. Aided in frontend and backend development.

NYC Department of Health and Mental Hygiene **Queens, NY**
Information Technology Specialist Intern *June 2024 - Jan. 2026*

- Modernized facility workstations by deploying and configuring upgraded monitors, keyboards, and peripherals.
- Audited hardware inventory and decommissioned laptops for compliant e-waste and asset lifecycle management.
- Helped system-wide (**3k+ computers**) recovery after global server outage by CrowdStrike. Imaged **62 PCs**.

PASSION PROJECTS

SpringForge | *Python, VPython, Computational Geometry, Mechanical Kinematics, Calculus*

- Developed a Python/VPython simulator modeling rotational motion in configurable multi-spring systems.
- Implemented **nonlinear torque calculations** and **numerical integration** for real-time angular motion.
- Built a polygon editor that computes **center of mass** and **inertia** for **custom shapes** via shoelace corollaries.

Slipstream | *Java, Processing, Client-Server Networking*

- Engineered a Java/Processing multiplayer racer with real-time **client-server synchronization** and state replication.
- Built pixel-based collisions, scrolling camera logic, and terrain-aware vehicle physics.
- Developed client-server networking to **synchronize opponent movement** and game state across machines.
- Implemented drifting, nitro, lap timing, HUD, and audio sound effects.

Bik | *C, MIPS Assembly, CMake, Systems-Level Programming, Compiler/Interpreter Design*

- Built a C compiler translating a custom high-level language into **executable MIPS assembly** with code generation.
- Designed a lexer, **recursive-descent parser**, AST, and scoped variable environment for end-to-end compilation.
- Implemented compile-time optimizations, including **constant folding**, to improve generated-code efficiency.

WORKFORCE SKILLS

- **Languages:** English (fluent), Mandarin Chinese (proficient), & Spanish (intermediate)
- **Technical Skills:** C, C++, Java, Kotlin, Lua, Go, Javascript, Typescript, Python, CMake, CSS, HTML, Next.js, TailwindCSS, PostgreSQL, Prisma, MongoDB, Node.js, React.js, MaterialUI, etc.
- **Tools:** Visual Studio Code, Android Studio, Neovim, AWS, Google Cloud, Supabase, etc.